10. See, for example, “Reply to Criticisms” in *Albert Einstein,* vol. 7 of *The Library of* Living Philosophers, P. A. Schilpp, ed. (New York: MJF Books, 2001).

11. W. J. van Stockum, Proc. R. Soc. Edin. A 57 (1937), 135.

12. The expert reader will recognize that I am simplifying. In 1966, Robert Geroch, who was a student of John Wheeler, showed that it is at least possible, in principle, to construct a wormhole without ripping space. But unlike the more intuitive, space-tearing approach to building wormholes in which the mere existence of the wormhole does not entail time travel, in Geroch’s approach the construction phase itself would necessarily require that time become so distorted that one could freely travel backward and forward in time (but no farther back than the initiation of the construction itself).

13. Roughly speaking, if you passed through a region containing such exotic matter at nearly the speed of light and took the average of all your measurements of the energy density you detected, the answer you’d find would be negative. Physicists say that such exotic matter violates the so-called averaged weak energy condition.

14. The simplest realization of exotic matter comes from the vacuum fluctuations of the electromagnetic field between the parallel plates in the Casimir experiment, discussed in Chapter 12. Calculations show that the decrease in quantum fluctuations between the plates, relative to empty space, entails negative averaged energy density (as well as negative pressure).

15. For a pedagogical but technical account of wormholes, see Matt Visser, *LorentzianWormholes: From Einstein to Hawking* (New York: American Institute of Physics Press, 1996).

Chapter 16

1. For the mathematically inclined reader, recall from note 6 of Chapter 6 that entropy is defined as the *logarithm* of the number of rearrangements (or states), and that’s important to get the right answer in this example. When you join two Tupperware containers together, the